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Fracture Distribution and Occurrence of DNAPL in a Clayey Lodgement Till

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This study examines the spatial fracture network in a clayey lodgement till, and the preferential flow pattern of DNAPL in the till. The study was conducted on a former gasworks site contaminated with coal tar. Fracture analysis was carried out and four fracture systems were recognised. Apart from fractures, burrows and root channels were recorded in the near surface sediments. Field investigation shows that two fracture systems were formed subglacially by loading of a glacier and horizontal shear within the lodgement till. Two other fracture systems were formed subsequently by desiccation and unloading/freeze-thaw processes in the unsaturated zone. Spatial distribution of free-phase DNAPL in the lodgement till is controled by the fracture network. The migration path of the DNAPL through the fracture systems is predominantly vertical in the upper 2 m and horizontal between 2 and 3.5 m. b.s. Below this depth the migration pathway is vertical. DNAPL was visually observed to a depth of 9 m.b.s. within fractures which continued downwards to unknown depth. Upscaling of the fracture properties was performed and a regional fracture model is suggested as a tool in future remidiation.

Introduction

The study of groundwater flow and migration of pollutants in and through clayey till started in Denmark in the late 1980's (Fredericia 1987; Fredericia 1990; Fredericia *et al.* 1990; Haldorsen and Krüger 1990; Villholth *et al.* 1990) and it became evident that fractures and other macropores played a very important role. The importance of fractures was emphasized by field investigations (Fredericia 1991; Jørgensen and Fredericia 1992; Fredericia 1993; McKay *et al.* 1993) and through flow-tests in large undisturbed columns (Jørgensen and Foged 1994; Hindsby *et al.* 1996; Jørgensen *et al.* 1998).

In order to evaluate the importance of fractures in clayey till seen in relation to infiltration of the groundwater and pollutants to underlying aquifers, the important issue is whether the fractures penetrate through the aquitard. It has been stated that fractures in clayey till have not been observed deeper than 6 to 7 m.b.s. (Jørgensen *et al.* 1998a; Baumann and Foged 1998), although fractures penetrating as deep as 17 m.b.s. have been reported (Fredericia 1991; Fredericia 1993; Klint and Fredericia 1997).

Another important question is to what depth the fractures are hydraulically active, as they might close due to the increasing pressure with depth. In general there is a decrease of bulk hydraulic conductivity with depth. In the oxidized zone the bulk hydraulic conductivity typically ranges from 10^{-5} to 10^{-7} m/s and there is a large reduction in hydraulic conductivity going from oxidized till down to reduced till where it decreases to 10^{-9} m/s based on large undisturbed column measurements (Jørgensen *et al.* 1998b). This is in good agreement with the decrease of fracture bulk intensity seen in tills going from oxidized till to reduced till (Klint and Fredericia 1995; Klint 1998). Jørgensen and Fredericia (1992) demonstrated indirectly that the upper 10 m, out of a sequence of 15 m of clayey till, was dominated by fracture flow. To what extent the fractures contributes to the bulk hydraulic conductivity at deep levels has yet to be determined. However, numerical analysis of solute migration through a fractured aquitard has shown that small, widely spaced, fully penetrating vertical fractures can cause considerable degradation of water quality (Harrison *et al.* 1992)

Field investigations have so far mainly been performed within the upper 5 to 6 m for practical reasons. Remediation of a former gasworks site in Haslev, involved excavation of till to a depth of 9 m.b.s. The excavation in Haslev is very deep compared to depths examined in earlier fracture investigations, and it provided a good opportunity to study fractures in a clayey till, 3 to 4 m deeper than most earlier investigations. Besides, the Haslev site is polluted with coal tar, and the site provided a good opportunity to study the occurrence and distribution of coal tar (DNAPL) in a fractured clayey till.

The objective was to perform a fracture analysis at the site, and to correlate the distribution of pollution to the fracture system. Furthermore it was intended to make a conceptual fracture model which is valid for the whole area, and act as a model in future remediation projects. To do so the fracture properties must be closely correlated to the geological model, and the geology must be consistent for the area as is the case for the township of Haslev.

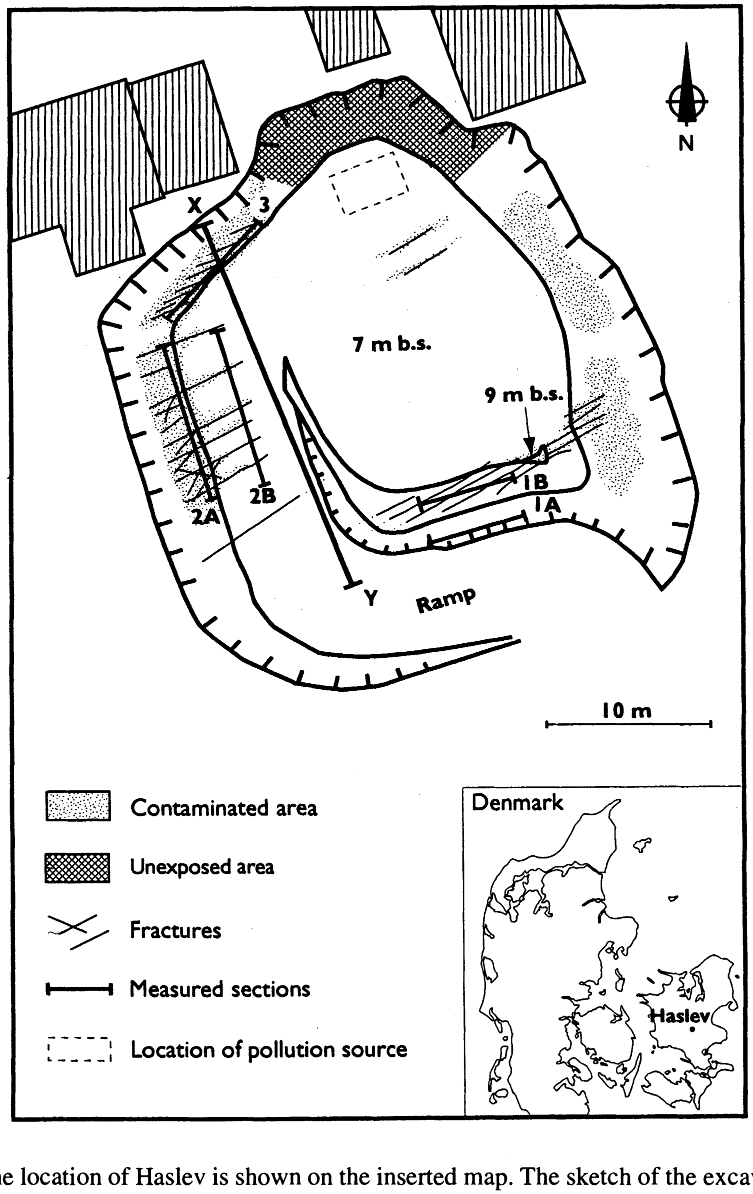
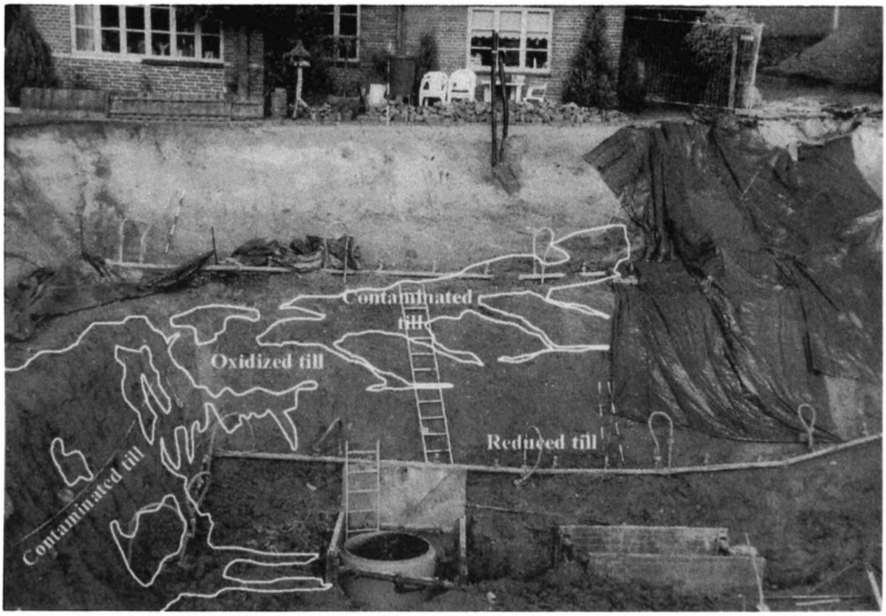


Fig. 1. The location of Haslev is shown on the inserted map. The sketch of the excavation on the gasworks site at Haslev shows the position of the investigated profiles. The walls in the pit were dipping 45 degrees. Profiles 1A and 1B were situated at levels 3 and 6 m.b.s., profiles 2A and 2B on the western wall at levels 3.5 and 6 m.b.s. and profile 3 on the northern wall at level 3 m.b.s. Profile Y-X is a complilation of profiles 1 to 3, and is represented in Fig. 6. The fracture trace of the most prominent vertical fractures is shown and the placing and distribution of the visually observed coal tar within the fractures is also indicated.



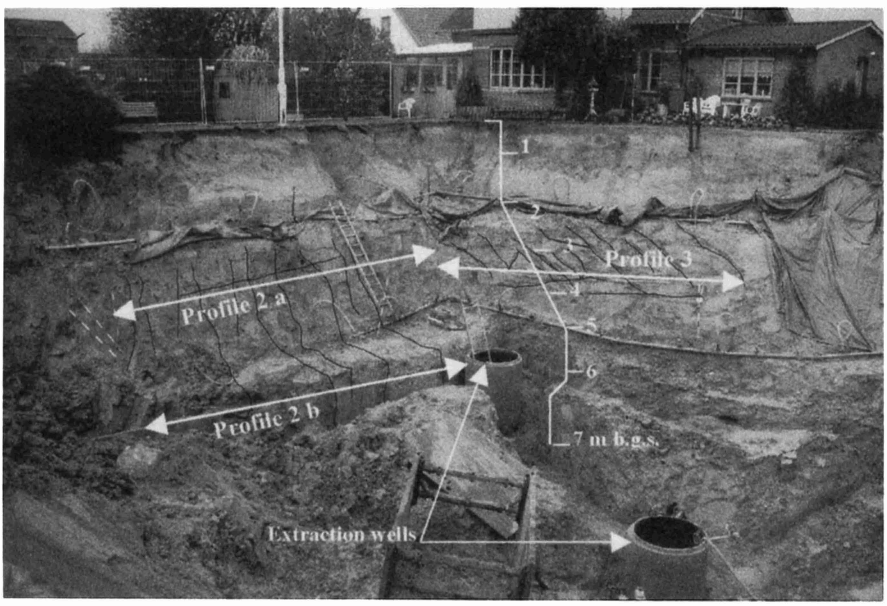


Fig. 2. The upper photo shows the distribution of coal tar in the northwestern part of the pit. The lower photo shows the position of profiles 2a, 2b and 3. The bottom of the extraction wells were excavated to a maximum depth of 9 m b.g.s, in which depth a fracture with free-phase DNAPL was observed.

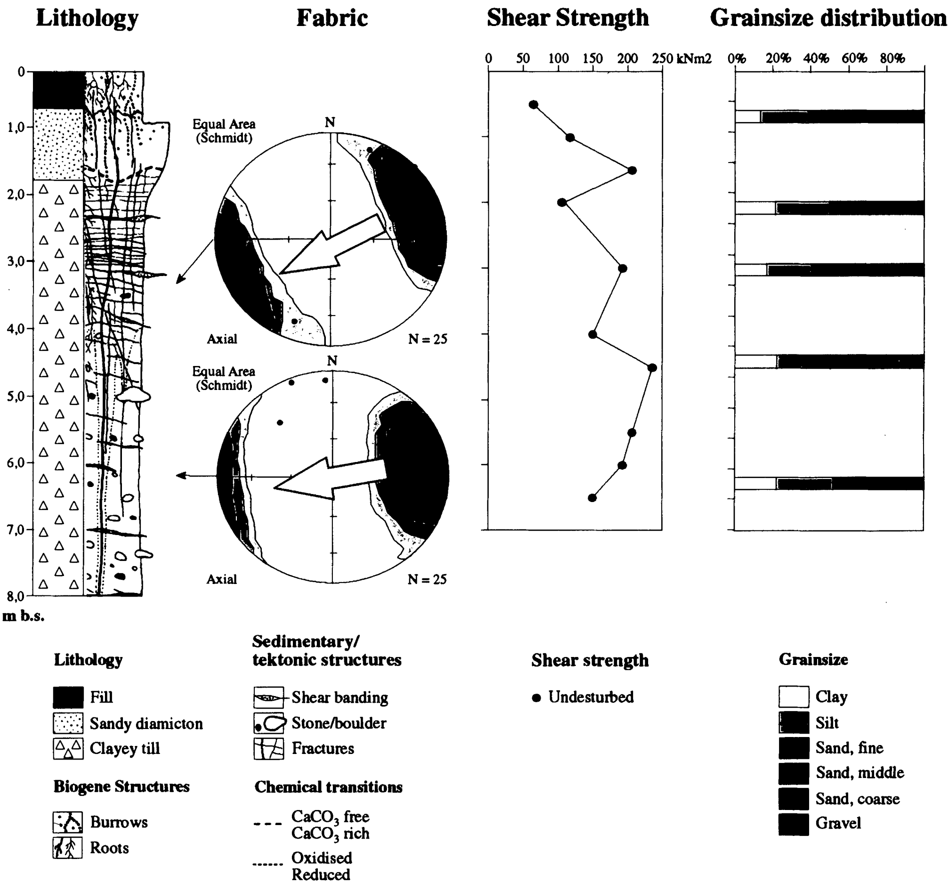


Fig. 3. Lithological and geotechnical parameters of the units exposed in the excavation illustrated as lithological units, clast fabric, shear strength and grain size distribution.

Description of Locality and Method

The locality is a former gasworks site located in the town of Haslev (Fig. 1). The gasworks was in use from 1899 to 1965 The contaminated soils were excavated in 1997 creating a temporary excavation in which fracture mapping was conducted (Fig. 2). The excavation has a ledge at 5 m.b.s. and trenches was dug to 7 m.b.s. The deepest part is reached at 9 m.b.s.

Fracture parameters were measured along 5 profiles on three walls in the excavation (Fig. 1). The lithological log of the sequence seen in the excavation (Fig. 3) was measured on the western wall across profiles 2A and 2B. The colours mentioned, refer to the Munsell soil colour chart (Munsell Color 1994).

In situ measurements of geotechnical parameters were made to estimate whether the till has been consolidated by ice loading and thereby indicates whether it is a sub-glacial till or not. Clast fabric measurements where made in order to characterize the depositional environment of the clayey till *i.e.* whether it is a lodgement till, ablation till or flow till. Fabric is furthermore used to interpretate the ice-movement direction. Knowledge of the type of till in which the fractures are developed is very important in interpreting the origin of the fractures as well as in the estimation of the regional extension of the fracture systems.

The smeared clay was cleaned off at the excavation walls and the fractures were carefully exposed, with knives or trowels, and then measured along a horizontal scan-line. The fracture order, position, size (trace length and depth), shape, orientation, surface texture and precipitation on all major fractures (typical with fracture surface area larger than 0.25 m^2) were measured and described according to Klint and Fredericia (1995, 1998). The presence and distribution of coal-tar was mapped by visual observation during the fracture measurement. The following fracture parameters were calculated from these data:

Fracture Orientation was used to classify fractures into distinct fracture sets or systems with characteristic properties. The orientation is illustrated on a stereographic projection, where all fractures are plotted as the pole to the fracture plane, or in a Rose Diagram showing the strike variation of the fractures. The latter method gives a visual presentation of the fracture orientation, but it does not account for the dip and is only used with vertical/sub-vertical fracture data.

Fracture Spacing is the mean perpendicular distance between adjacent fractures in a parallel fracture set. It was measured directly in the field along the different profiles, or calculated from the length of the scan-line, orientation and number of fractures in one system (Klint and Fredericia 1998).

Fracture Intensity is the inverse of the fracture spacing (number of fractures pr m). The fracture intensity allows the spacing to be illustrated in a rose diagram or a graph. Fracture intensity becomes larger with increasing number of fractures.

Fracture Trace Frequency illustrates the spatial fracture distribution and variation both laterally and vertically. It must be noted that the frequency of fractures nearly parallel with the wall are biased and the average numbers are typical 1.3-2 times smaller than the true Fracture Intensity.

Geological Setting

The town of Haslev is situated on a ground moraine plain which continues to the west. The plain has a drumlinoid morphology with large scale east-west oriented swells and sags. Lithologically the plain consists of clayey till, which in the sags is covered with late-glacial and post-glacial alluvial deposits, outlining the elongated morphology of the plain. Borehole logs from the Haslev area show that the thickness of the clayey till is between 10 and 20 m (Binzer 1977). The till is usually resting on Palaeocene limestone, but in some borings a layer of glacio-fluvial sediments is present between the till and the limestone. The site next to the former gasworks site is the waterworks for the town of Haslev. Borings from the waterworks site show, that there is between 15 and 17 m of glacial deposits next to the investigated site. In two of the borings a 1 to 2 m thick layer of glacio-fluvial sediments is recorded between the clayey till and the Palaeocene limestone. Groundwater levels were measured in November 1993 during an investigation of the contamination at the site (Krüger 1994) and the water table was between 1 and 2 m.b.s. Filters at 10 m.b.s. had the same head as the underlying limestone aquifer, which is between 4 and 5 m.b.s.

Lithological Description and Classification

The uppermost 0.4 m consist of an organic rich soil, in some places mixed with coal tar. The soil is strongly bioturbated and with many roots. In places the uppermost part of the sequence consits of fill.

From 0.4 to 1.2 m.b.s. a silty and clayey sandy diamicton is recorded with clay streaks and small amount of gravel and few stones. It is non-calcareous, light olive brown with reddish brown and light grey spots. Cryoturbation is seen in this unit, indicating periglacial remoulding. Roots are abundant and always associated with the light grey spots. Worm holes with a diameter of 3 to 6 mm are abundant in this unit. The worm holes are open and some have a coating of clay and organic matter.

From 1.2 m.b.s. and downwards a clayey till with small amounts of gravel, few stones and occasional subhorizontal streaks of sand or silt is recorded. The uppermost 0.5 m of the unit is quite sandy and it is non calcareous to 1.4 m.b.s. The colour is olive brown with reddish brown and light grey spots down to about 4.5 m.b.s. and olive grey below. Along major vertical fractures the olive brown colour is recorded down to 8 m.b.s. in the southern part of the excavation. Between 2 and 3.5 m there are laterally extensive areas with a dark olive grey colouring within the olive brown unit, mainly associated with intense horizontal fracturing.

There are many roots down to about 1.75 m.b.s. which are always associated with the light grey spots and fracture surfaces. Worm holes are recorded in the non-calcareous part, down to about 1.4 m. They are 3 to 6 mm wide. They where not counted, but roughly estimated there is about 100 pr m^2.

Shear vane strength was measured in the sandy diamicton and the clayey till in the interval between 0.5 and 6.5 m.b.s. (Fig. 3). The sandy diamicton shows shear vane strengths between 60 and 100 kN/m2. The shear vane strength of the clayey till is between 140 and 230 kN/m^2, with one exception at 2 m.b.s. where the shear vane strength is 100 kN/m2. The low value in the clayey till is obtained at a level with very high intensity of horizontal fractures.

Clast fabric measurements where made on the clayey till at 3 m.b.s. and 7 m.b.s. At 3 m.b.s. the average clast fabric orientation is 65° and at 7 m.b.s. the average clast fabric orientation is 86° (Fig. 2). Both fabric measurements are unimodal and significant, with eigen values of 0.820 and 0.806. A few measurements of scour marks on *in situ* clasts show an orientation between 65° and 85°. The direction of movement of the ice sheet which deposited the clayey till is interpreted to be from east-northeast to west-southwest. This direction corresponds well with the 'drumlinoid' morphology west of Haslev.

The shear vane strength, the strong fabric within the clayey till and the scour marks on clasts indicates that the clayey till is a lodgement till, deposited underneath an ice sheet, moving from east-northeast towards west-southwest.

Fractures

Fracture Description

Fractures in the lodgement till were described and classified according to Klint and Fredericia (1998). The fractures are divided into four systems based on their orientation and surface characteristics. The fracture parameters are shown in Fig. 3 and a conceptual geological model of the site, comprising the fracture parameters and lithological data, is shown in Fig. 4.

The system 1 fractures are large vertical/subvertical fractures, and they are the most prominent fractures recorded at the site. The strike of the system 1 fractures is 65° (Fig. 4). These fractures are straight, planar and with a rough surface. The spacing of the system 1 fractures varies throughout the excavation and it is about 60 cm in profiles 1A and 1B. At profile 1 the fractures are recorded to the bottom of the excavation, which is 9 m.b.s. Several of the system 1 fractures in profile 1 have a halo of oxidized till, 5 to 10 cm wide, down to 8 m.b.s. On the southern wall in the excavation the system 1 fractures have a lateral extent of at least 20 m (Fig. 1). Along profile 2, the spacing varies along the 11 m long profile, from 50 cm in the southern part of profile 2A and 2B, to 1,5 m in the northern part. Along profile 2B, system 1 fractures have been observed to the bottom of the excavation and are thus traced to 7 m.b.s. in this part of the excavation. Along profile 3 the spacing is nearly uniform with an average spacing of 60 cm. However, along profile 3 the system 1 fractures terminate at 5 m.b.s. where they meet a subhorizontal sand layer about 2 cm thick.

The system 2 fractures are horizontal/subhorizontal fractures which are present throughout the sequence. They are typical 1 mm wide and mostly filled with silt or fine sand and they are dipping slightly towards the east-northeast. The horizontal trace length of the system 2 fractures is more than 6 m and the average spacing is 80 cm.

The system 3 fractures are restricted to the uppermost 4.5 m and consist of numerous small horizontal fractures. The horizontal trace length of the fractures is 5 to 20 cm. The spacing of the fractures in the system varies considerably. In general the intensity of the system 3 fractures decreases with depth and they terminate at about

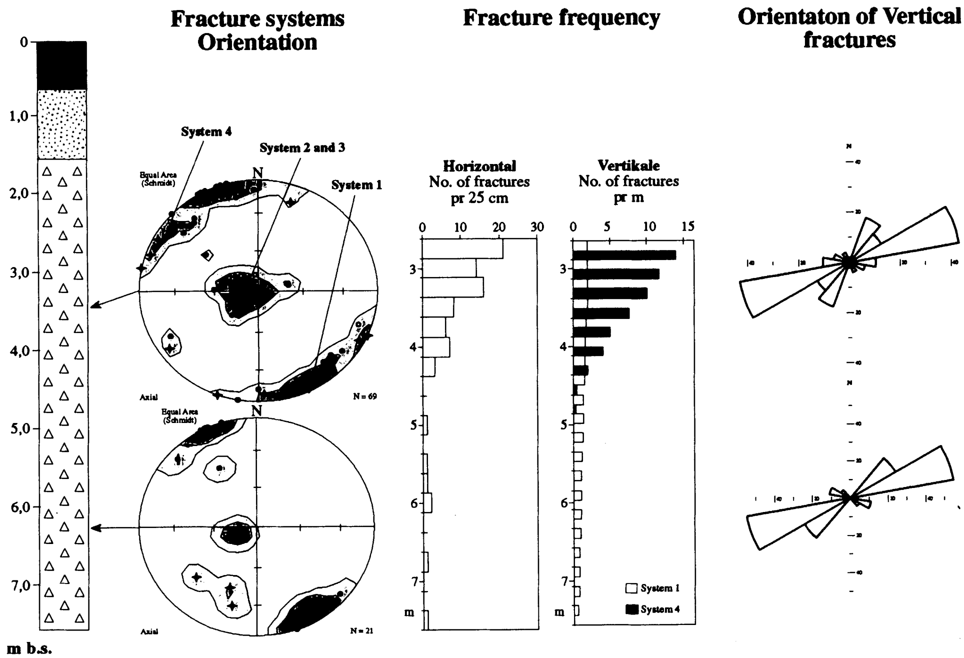


Fig. 4. Outline of fracture parameters of the observed fractures illustrated as systems, orientation, fracture trace frequency, and orientation and distribution of vertical fratures.

4.5 m. The spacing is 1 to 1.5 cm between 2.5 and 3 m.b.s., and 4 cm at 4 m.b.s. In places there are zones with higher intensity and thus smaller spacing.

The system 4 fractures are vertical/subvertical fractures. They have an irregular and undulating shape and a more random orientation than the system 1 fractures. There is a small concentration between 30° and 50°. The fracture trace frequency is highest in the upper part of the lodgement till and decreases downward with depth to where the fractures terminate at about 4.5 m.b.s.

Classification of Fractures

The system 1 fractures have characteristics of glaciotectonic fractures (Klint and Fredricia 1997). They are very large, occur in the lodgement till, are straight and have a systematic orientation parallel to the direction of ice movement. They occur both in the oxidized till as well as in the reduced till (above and below lowest groundwater table) and cannot be regarded as desiccation fractures. They are interpreted as glaciotectonic fractures, caused by the load of an overriding glacier. Fracture patterns of glaciotectonic fractures parallel with the ice movement direction is previously described in lodgement till (Fredericia 1987; Fredericia 1991; Klint and Fredericia 1995). Fractures perpendicular to the ice movement direction also occur

(Jørgensen and Fredericia 1992; Klint and Fredericia 1997; Klint and Gravesen 1998).

The large subhorizontal fractures belonging to the system 2 fractures dip weakly towards the direction of ice movement (Fig. 4) as determined by clast fabric analysis (Fig. 3). They have a fairly constant spacing and are mostly filled with silt or fine sand. They are interpreted as shear fractures produced by subglacial shear within the clayey lodgement till. The silty filling is presumably former silt-lenses, sheared out in the shear process, which also created the clast fabric within the lodgement till.

The smaller horizontal fractures belonging to the system 3 fractures are restricted to the upper 4.5 m which is the oxidised clayey till. The near surface position and the increase of intensity towards the ground surface suggest that the system 3 fractures are created by freeze-thaw cycles. They may initially have been formed as pressure release fractures during glacial unloading, and/or subglacial shear and later enhanced by freeze and thaw processes.

The system 4 fractures occur in the oxidized zone above the lowest groundwater table. They are irregular and undulating, and they have a more random orientation than the glaciotectonic fractures and are therefore interpreted as desiccation fractures. Typically desiccation processes reactivate the glaciotectonic fractures and form a set of desiccation fractures perpendicular to those (Klint and Fredericia 1995).

Occurrence of DNAPL

Free-phase coal tar was observed only within the fractures in the clayey till. There is, however, a pronounced smell of tar in the matrix indicating that the aromatic compounds from the DNAPL have migrated into the matrix probably in a dissolved phase. The coal tar was liquid and highly viscous and it flowed freely out of the opened fractures at all levels where it was observed.

The free-phase DNAPL was observed in all four fracture systems. In the system 1 fractures DNAPL was observed in all the examined profiles. Free-phase DNAPL was observed in fractures from near ground surface to the bottom of excavation (9 m.b.s. for profile 1B and 6.5 m.b.s. for profile 2B). Along profile 2B about 20% of the glaciotectonic fractures contained free-phase DNAPL below 5 m.b.s. All system 1 fractures along profile 3 contained DNAPL down to 5 m.b.s. where the fractures ceased. Some of the glaciotectonic fractures had no halo or other indications of hydraulic activity below 4.5 m, but they where nevertheless conducting free-phase DNAPL (Fig. 5). Other fractures had a halo of oxidized till as deep as 8 m.b.s. and some of those did not conduct free-phase DNAPL. In the uppermost 4.5 m many of the systems 2 and 3 fractures and most of the system 4 fractures contained free-phase DNAPL.

Fig. 5. Glaciotectonic fracture in clayey lodgement till 5 to 6.5 m.b.s. containing DNAPL along profile 2B.

Migration Path of DNAPL

In the excavation the free-phase DNAPL migrated preferential in the spatially distributed fracture systems. It has migrated from the contamination source vertically down through the uppermost 1.5 to 2 m of fill, sandy diamicton, worm holes and desiccation fractures. At about 2 m.b.s. the DNAPL migrated into the horizontal fractures. The zones of intense horizontal fracturing (system 3 fractures) play a very important role in the spreading pattern of the DNAPL . In the intensely fractured zones it migrated laterally and the horizontal extent of the contamination is about 20 m from the source (Fig. 1). The system 3 fractures are only about 10 to 20 cm long, but the high frequency results in a high connectivity of the fractures.

The DNAPL migrated from the horizontal fractures into the vertical system 1 fractures. In the large vertical fractures the DNAPL has migrated downwards. The DNAPL is spatially distributed in an array of parallel fractures and it is observed down to a depth of 9 m.b.s. within a system 1 fracture in the deepest part of the excavation. These fractures continue to unknown depth below the bottom of the excavation. Aromatic compounds of the DNAPL in form of benzene, ethylbenzene, toluene and xylene have been detected in near by water wells (Krüger 1994), indicating that at least dissolved compounds of the DNAPL have reached the Palaeocene limestone which is the primary groundwater aquifer in this region. Free-phase DNAPL might also have reached the aquifer.

Discussion

The fractures reaching the deepest levels at Haslev are glaciotectonic fractures and they are visually recorded to a depth of 9 m.b.s., and continue downward to unknown depth. The lateral extent of the glaciotectonic fractures is more than 20 m. It is therefore reasonable to assume that some of the vertical fractures might continue through the clayey till which is 15 to 17 m thick, or at least almost through the unit. Some of the glaciotectonic fractures where acting as preferential flow paths of free-phase DNAPL to at least 9 m.b.s. If the fractures continue through the clayey till, it is very likely that they have conducted free-phase coal tar into the primary aquifer. In two out of four nearby water wells aromatic compounds were detected (Krüger 1994), indicating that at least dissolved compounds from the DNAPL were transported to the aquifer. Whether or not the fractures continue all the way through the clayey till, the presence of aromatic compounds in the aquifer shows that even 15 to 17 m of clayey till does not act as a protective aquitard, and the preferential flow in spatially distributed fractures play an active and important role.

The DNAPL migrates horizontally up to 20 m from the contamination source in horizontal fractures 2 to 4 m.b.s., despite that DNAPL is drained downwards as well through the glaciotectonic fractures. Gravity controlled downward migration would in the first place be expected. An explanation could be that the flux of coal tar into the soil exceeds the DNAPL conductivity of the relatively few vertical fractures. The system 3 fractures on the other hand are small but numerous and they have a high fracture porosity and connectivity. Consequently, there is a preferential horizontal flow which creates a widely distributed shallow contamination plume.

Based on the method described by Klint and Fredericia (1998) an accurate fracture model, which describes the actual fracture parameters as well as their distribution and variability has been established at the site. When addressing the issue of upscaling of the fracture properties, characterisation of the geological setting of the fracture network is required. The system 1 and system 2 fractures are related to internal shear and ice load in a lodgement till. The whole area is covered by the lodge-

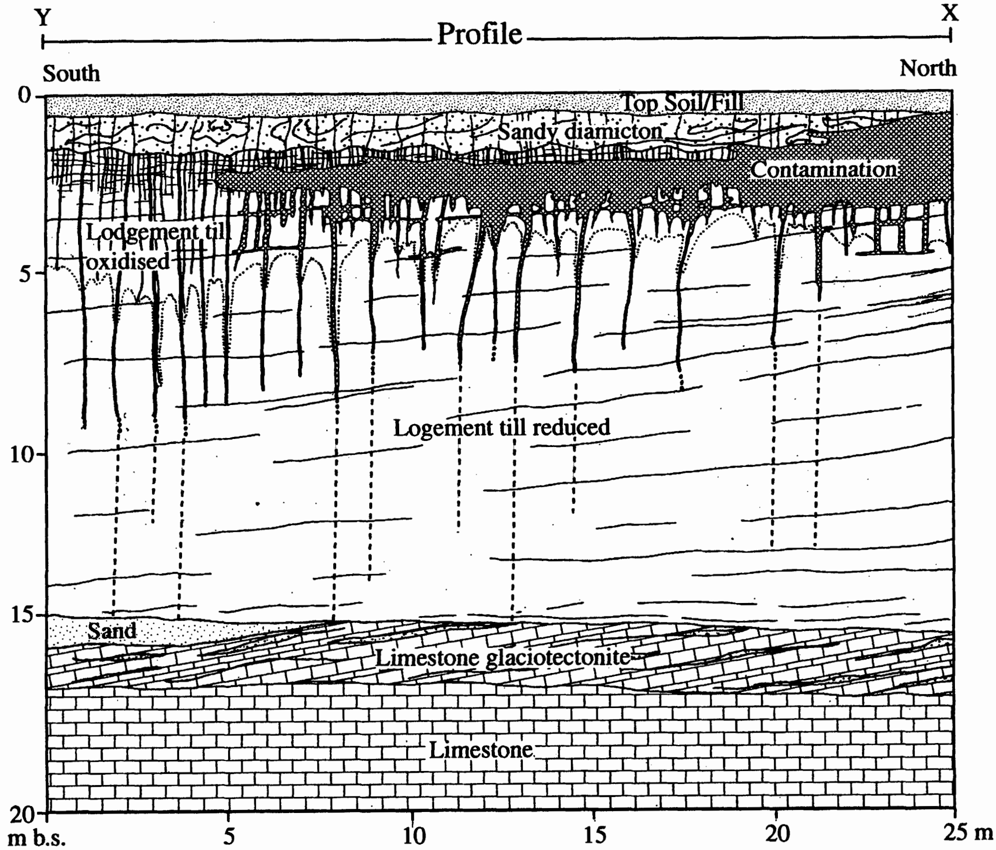


Fig. 6. Conceptual model of the geology and the fracture distribution at the investigated site (profile Y-X in Fig. 1). The inferred continuation of the vertical fractures, below the lowest level of the excavation is shown with broken lines. The fractures containing DNAPL are indicated. Between 2 to 4 m the contamination plume is indicated more than the single fractures containing DNAPL.

ment till and these two fracture systems must therefore be expected to be present regionally. The near surface system 3 and 4 fractures are related to desiccation and freeze-thaw processes that has occurred regionally as well. The conceptual model for the site (Fig. 6) shows that both the depth and the spacing of the glaciotectonic fractures vary along the 25 m long profile. Taken the length of the profile it is very likely that spatially variations in fracture systems is well represented. As the geological setting is very consistent in the area, it is very likely that the fracture model is valid for the whole township and surroundings. This makes the conceptual fracture model a very useful and import tool in pollution risk assessment and remediation strategies within the whole township and surroundings.

Conclusions

Fracture analysis has been carried out on a clayey lodgement till and a complex spatially distributed fracture network is revealed. The network consists of four fracture systems; two vertical and two horizontal ones. Apart from fractures, burrows and root channels are recorded in the near surface sediments. The largest vertical fractures are visually observed to a depth of 9 m.b.s. and continuing downwards to unknown depth.

The fracture systems are created by different processes; loading and movement of an overriding glacier, shear within the till, desiccation, and freeze-thaw cycles. A conceptual fracture model is established for the site and it is believed to be valid for the township of Haslev and surroundings.

The spatially distributed fracture network degrade the protective properties of the clayey lodgement till. Free-phase DNAPL occurs within the clayey lodgement till in fractures and there is a preferential flow of free-phase DNAPL in the spatially distributed fracture network. Both the number of fractures and the presence of free-phase DNAPL is highest in the upper 3 to 4 m. The migration path of the free-phase DNAPL through the fracture systems is predominantly vertical in the upper 2 m, horizontal between 2 and 3.5 m.b.s. and vertical below. The presence of free-phase DNAPL within the fractures shows that they have acted as pathways through the clayey till to at least 9 m.b.s.

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References

Baumann, J., and Foged, N. (1998) Fractures in a geotechnical perspective. Abstract in: Mass Transport in Fractured Aquifers and Aquitards. Conference at Geoscience Center Copenhagen, May 14-16 1998.

Binzer, K. (1977) Geological Basic Data Map, 1512 IV Næstved, 1:50.000. DGU.

Fredericia, J. (1993) Quality and quantity of groundwater recharge – dependency on textural conditions. Groundwater recharge – Quality and quantity (in Danish). ATV- Committee concerning groundwater polution. May 4, 1993.

Fredericia, J. (1991) Fractures in Clayey Till. Occurence, Genesis and Hydrogeological Significance. PhD-Thesis. Technical University of Denmark, Institute for Geology and Geotechnics. (English resume).

Fredericia, J., Jensen, P., Lind, B., Lundin, L., and Nyborg, M. (1990) Saturated hydraulic conductivity. Chapter 2,2796, in: Haldorsen, S. 1990: Hydrogeological properties of Nordic tills. Nordic Hydrogeological Programme NHP, Report No. 25, 250 pp.

Fredericia, J. (1990) Saturated hydraulic conductivity of clayey till and the role of fractures, *Nordic Hydrology, Vol. 21*, pp. 119-132.

Fredericia, J. (1987) Fractures and permeability in clayey till. ATV-meeting, Okt. 29. 1987, 1-28 (in Danish).

Haldorsen, S., and Krüger, J. (1990) Till genesis and hydrological properties, *Nordic Hydrology, Vol. 21*, pp. 81-91.

Harrison, B., Sudicky, E.A., and Cherry, J.A. (1992) Numerical analysis of solute migration through fractured clayey deposits into underlying aquifers, *Water Resour. Res. Vol. 28, No 2*, pp 515-526.

Hindsby, K., McKay, L.D., Jørgensen, P., Lenczewski, M., and Gerba, C.P. (1996) Fracture aperture measurement and migration of solutes, viruses, and immiscible creosote in a column of clay-rich till, *Ground Water, Vol.34, no. 6*, pp. 1064-1075.

Jørgensen, P., and Foged, N. (1994) Pesticide leaching in intact blocks of clayey till. Thirteenth Int. Conference on Soil Mechanics and Foundation Engeneering, New Delhi, India pp. 1661-1664.

Jørgensen, P.R., Baumann, J., Helstrup, T., Urup, J., and Butzbach, K. (1998a) Hydraulic properties in fractured clayey till – characteristics of 8 Danish localities with clayey till, *Geologisk Nyt* 2/98, pp. 13-16.

Jørgensen, P.R., and Fredericia, J. (1992) Migration of nutrients, pesticides and heavy metals in fractured clayey till, *Geotechnique, Vol. 42*, pp. 67-77.

Jørgensen, P., McKay, L.D., and Splid, N.H. (1998b) Evaluation of chloride and pesticide transport in a fractured clayey till using large undisturbed columns and numerical modeling, *Water Resour. Res., Vol. 34 No. 4*, pp. 539-553.

Krüger AS (1994) Supplementary contamination investigations on the Haslev Gasworks site. Report (in Danish).

Klint, K.E.S. (1998) Fractures in clayey tills at Flakkebjerg, Denmark "Origin and distribution" Abstract Nordic Geological Vinter-meeting, University of Århus, Jan. 1998.

Klint, K.E.S., and Fredericia, J. (1998) Quantitative fracture characterisation in clayey diamict sediments. "A recipe". Extended abstract in: Mass Transport in Fractured Aquifers and Aquitards. Conference at Geoscience Center Copenhagen, May 14-16, 1998, pp. 4.

Klint, K.E.S., and Fredericia, J. (1997) Fractures in clayey till, origin and extention, Groundwater research in Denmark 1992-96, ATV- Committee concerning groundwater polution. April 23 1997. Extended abstract. Schæffergården, 23 april 1997, pp. 67-84.

Klint, K.E.S., and Fredericia, J. (1995) Fracture parameters in clayey till *Vand & Jord*, oktober, pp. 208-214.

Klint, K.E.S., and Gravesen, P. (1998) Fractures and biopores in weichselian clayey till aquitards at Flakkebjerg, Denmark. Abstract in: Mass Transport in Fractured Aquifers and Aquitards. Conference at Geoscience Center Copenhagen, May 14-16.

McKay, L.D., Hindsby, K., Morthorst, J., and Lenezewski, M. (1993) Field and laboratory tracer experiments in fractured clayey till in Denmark. In: Geological Society of America, 1993 Annual Meeting, Boston, Massachusetts, October 25-28. A-427.

Munsell Collor (1994) *Munsell soil color charts*. 1994 revised edition, Mackbeth Division of Kollmorgon Instruments Corporation.

Villholth, K., Jensen, K.H. and Fredericia, J. (1990) Field investigations of preferential flow behaviour. NPo-report B 19, 32 pp.

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